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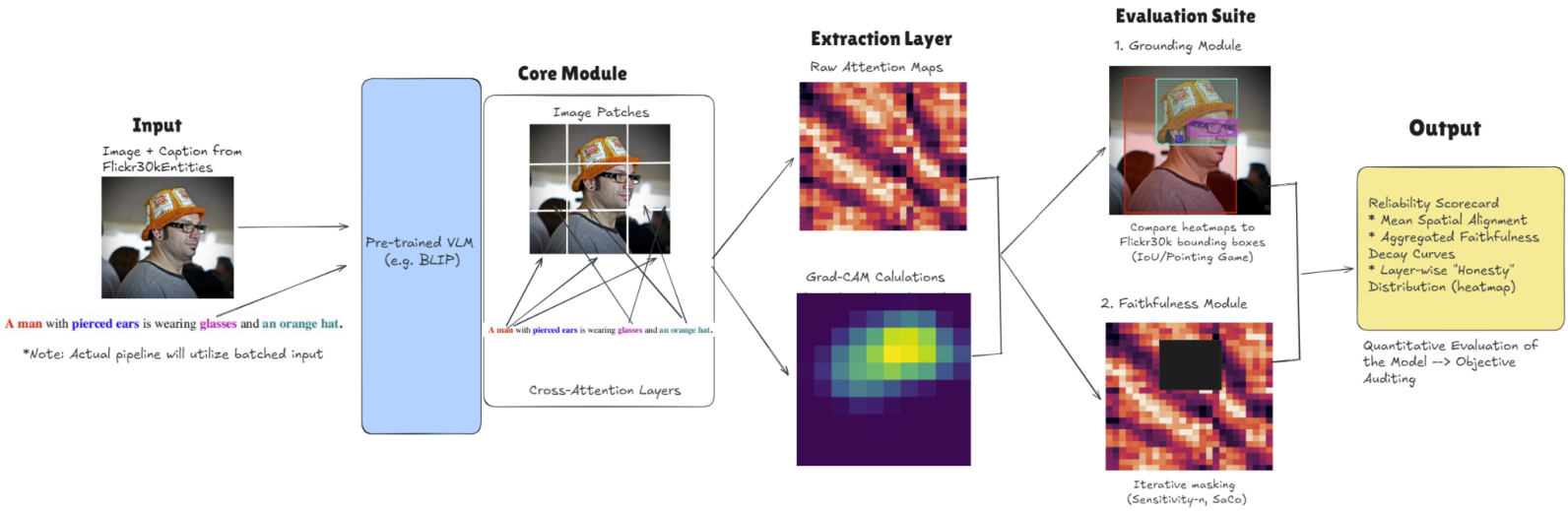
Group 9 Project Proposal

Title: VLM-Audit

Motivation

While Modern Vision-Language Models (VLMs) have achieved remarkable performance in tasks like image captioning, they remain largely "black boxes." A critical component of these models is the cross-attention mechanism, which governs how specific words "attend" to specific image regions. However, recent research suggests that raw attention weights are often uninterpretable or misleading, failing to represent the actual causal rationale behind a model's output. This lack of transparency is a significant hurdle for Explainable AI (XAI), especially in high-stakes deployments where "trustworthy AI" is mandatory. This project seeks to bridge this gap by building a diagnostic toolkit that moves beyond qualitative "heat map" visualizations. By quantitatively evaluating the faithfulness of these maps we can provide a rigorous framework for auditing multimodal system behavior.

Diagram



Evaluation

To evaluate the project, we will implement a two-fold quantitative strategy:

1. Grounding Accuracy: We will use the Flickr30k Entities dataset, which provides bounding boxes for specific phrases. We will calculate the Pointing Game Accuracy and Intersection-over-Union (IoU) between the generated heatmaps (both raw attention and Grad-CAM) and these manual annotations. This measures how "human-aligned" the model's focus is.

2. Faithfulness Metrics: To verify if the model's internal attention maps are truly representative of its decision-making process, we will perform masking analysis. This involves systematically masking the pixels identified as 'most important' by the model. A faithful explanation is one where the removal of these key pixels results in a significant and immediate drop in the model's prediction confidence, proving that the highlighted regions were indeed critical to the output (example metrics include Sensitivity-n, SaCo, etc).

We will measure Grounding and Faithfulness across selected subset of transformer layers. This allows us to identify where semantic alignment begins to emerge and which layers provide the most reliable explanations of model behavior.